Oracle GoldenGate - Initial Load & CDC Setup Made Easy with AdminClient

This article demonstrates the configuration of Initial Load and Change Data Capture (CDC) in Oracle GoldenGate 23.4 using the AdminClient command-line utility—without relying on the OGG Web Console.

It covers a 19c (19.27) Oracle Database environment on Oracle Linux 8.8, detailing both source and target pluggable database setups. Key highlights include process naming conventions for Extract, Replicat, and Distribution Path, as well as standard trail file naming formats for IL and CDC. A step-by-step approach with executed commands and logs ensures clarity, enabling seamless replication setup.

Database Environment:

```
Database Version : 19.27
O.S Version      : Oracle Linux 8.8
OGG Version      : 23.4 for Oracle on Linux(x86-64)
Source DB Name   : ORCL as CDB and ORCLPDB1 as Pluggable Database on vm1 Host.
Target DB Name   : REPLICA as CDB and RPDB1 as Pluggable Database on vm2 Host.
CMDLINE Utility  : AdminClient
```

GoldenGate Process Naming Examples

```
Extract Name for Initial Load      : EXT2_IL
Distribution Path For Initial Load : DSP2_IL
Replicat Name for Initial Load     : REP2_IL
Extract Name for CDC               : EXT2_CDC
Distribution Path for CDC          : DSP2_CDC
Replicat Name for Initial Load     : REP2_CDC
```

GoldenGate Trail File Naming Examples

```
Example: Initial Load -> il000000
Example: CDC          -> ex000000
```

Launching Oracle GoldenGate 23.4 AdminClient

```
[oracle@vm1 ~]$ cd $OGG_HOME/bin
[oracle@vm1 bin]$ ./adminclient
Oracle GoldenGate Administration Client for Oracle
Version 23.4.1.24.05 OGGCORE_23.4.0.0.0OGGR_LINUX.X64_240529.0500_FBO

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Linux, x64, 64bit (optimized) on May 29 2024 09:21:59
Operating system character set identified as UTF-8.
```

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Connecting to Oracle GoldenGate Deployment via AdminClient

```
OGG (not connected) 1> connect http://localhost:7802 user oggadmin password oggadmin
```

Viewing Credential Store and Logging into the Database in AdminClient

---

```
OGG (http://localhost:7802 dep1) 2> Info credentialstore
Default domain: OracleGoldenGate

  Alias: ORCL_SOURCE
  Userid: c##ggadmin@ORCL

Other domains:
  Network
To view other domains, use INFO CREDENTIALSTORE DOMAIN <domain>

OGG (http://localhost:7802 dep1) 3> DBLogin USerIDAlias ORCL_SOURCE
Successfully logged into database CDB$ROOT.
```

Checking Supplemental Logging and Replication Status for a Table

---

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 27> Info trandata
orclpdb1.scott.emp_e4
Logging of supplemental transaction log data is enabled for table ORCLPDB1.SCOTT.EMP_E4.
Logical Replication for table ORCLPDB1.SCOTT.EMP_E4 is enabled.
All columns supplementally logged for table ORCLPDB1.SCOTT.EMP_E4.
Prepared CSN for table ORCLPDB1.SCOTT.EMP_E4: 5773341
```

Create Initial Load Extract (EXT2_IL)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 7> add extract EXT2_IL
SOURCEISTABLE
2025-08-13T17:24:46Z  INFO    OGG-08100  Extract added.

OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 8> edit params EXT2_IL
2025-08-13T17:26:20Z  INFO    OGG-10183  Parameter file EXT2_IL.prm passed validity check.

OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 9> view params EXT2_IL
EXTRACT EXT2_IL
USERIDALIAS ORCL DOMAIN OracleGoldenGate
EXTFILE ma2ma/aug14/initial_load/il
Table ORCLPDB1.SCOTT.EMP_E4, SQLPREDICATE "AS OF SCN 5945163";
```

Create CDC Extract (EXT2_CDC)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 10> add extract EXT2_CDC
integrated tranlog begin now

2025-08-13T17:28:10Z  INFO    OGG-08100  Integrated Extract added.


OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 11> register extract EXT2_CDC
database container (orclpdb1)

2025-08-13T17:28:29Z  INFO    OGG-02003  Extract group EXT2_CDC successfully registered with
database at SCN 5907807.


OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 12> add exttrail
ma2ma/aug14/cdc/ex, extract EXT2_CDC

2025-08-13T17:28:45Z  INFO    OGG-12029  The item type file with name 'EXT2_CDC.prm' does not
exist.


OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 13> edit params EXT2_CDC
2025-08-13T17:32:44Z  INFO    OGG-10183  Parameter file EXT2_CDC.prm passed validity check.

OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 14> view params EXT2_CDC
EXTRACT EXT2_CDC
USERIDALIAS ORCL DOMAIN OracleGoldenGate
EXTTRAIL ma2ma/aug14/cdc/ex
Table ORCLPDB1.SCOTT.EMP_E4;
```

Check Status of EXTRACT Processes on Source

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 15> info all tasks
Program     Status      Group       Type              Lag at Chkpt       Time Since Chkpt
EXTRACT     STOPPED     EXT2_IL     SOURCE EXTRACT        00:00:00           00:00:00

OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 16> info all
Program     Status      Group       Type              Lag at Chkpt       Time Since Chkpt
EXTRACT     STOPPED     EXT2_CDC    INTEGRATED            00:00:00           00:08:06
```

ADD DISTRIBUTION PATH FOR INITIAL LOAD (DSP2_IL)

Note -> If you use the ogg protocol instead of wss, then you must use the TARGETTYPE option. The syntax in that case would be following below.

```
ADD DISTPATH path-name SOURCE source-uri TARGET target-uri [ TARGETTYPE ( MANAGER | COLLECTOR |
RECVSRVR ) ]

OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 19> ADD DISTPATH DSP2_IL SOURCE
trail://localhost:7803/services/v2/sources?trail=ma2ma/aug14/initial_load/il TARGET
ogg://vm2:7804/services/v2/targets?trail=ma2ma/aug14/initial_load/il TARGETTYPE RECVSRVR
```

The syntax for specifying authentication method using URI credentials is deprecated.

```
2025-08-13T17:39:50Z  INFO    OGG-08511  The path 'DSP2_IL' has been added.
```

ADD DISTRIBUTION PATH FOR CDC (DSP2_CDC)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 20> ADD DISTPATH DSP2_CDC SOURCE
trail://localhost:7803/services/v2/sources?trail=ma2ma/aug14/cdc/ex TARGET
ogg://vm2:7804/services/v2/targets?trail=ma2ma/aug14/cdc/ex TARGETTYPE RECVSRVR

The syntax for specifying authentication method using URI credentials is deprecated.
2025-08-13T17:40:53Z  INFO    OGG-08511  The path 'DSP2_CDC' has been added.
```

Check Status of Distribution Path on Source

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 21> Info DISTPATH DSP2_IL

 Path Name: DSP2_IL
 Status: stopped
 Source URI: trail://localhost:7803/services/v2/sources?trail=ma2ma/aug14/initial_load/il
 Target URI: ogg://vm2:7804/services/v2/targets?trail=ma2ma/aug14/initial_load/il
 Encryption Profile: LocalWallet


OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 22> Info DISTPATH DSP2_CDC

 Path Name: DSP2_CDC
 Status: stopped
 Source URI: trail://localhost:7803/services/v2/sources?trail=ma2ma/aug14/cdc/ex
 Target URI: ogg://vm2:7804/services/v2/targets?trail=ma2ma/aug14/cdc/ex
 Encryption Profile: LocalWallet
```

Create Initial Load Replicat (REP2_IL)

```
OGG (http://localhost:7802 dep1) 3> add replicat REP2_IL EXTTRAIL  ma2ma/aug14/initial_load/il
CHECKPOINTTABLE c##ggadmin.ckpt
2025-08-13T18:02:09Z  INFO    OGG-08100  Replicat added.

OGG (http://localhost:7802 dep1) 4> edit params REP2_IL
2025-08-13T18:02:37Z  INFO    OGG-10183  Parameter file REP2_IL.prm passed validity check.

OGG (http://localhost:7802 dep1) 5> view params REP2_IL
REPLICAT REP2_IL
USERIDALIAS RPDB1 DOMAIN OracleGoldenGate
REPERROR default, discard
MAP ORCLPDB1.SCOTT.EMP_E4, TARGET RPDB1.SCOTT.EMP_E4;
```

Create CDC Replicat (REP2_CDC)

```
OGG (http://localhost:7802 dep1) 6> add replicat REP2_CDC EXTTRAIL  ma2ma/aug14/cdc/ex
CHECKPOINTTABLE c##ggadmin.ckpt
2025-08-13T18:03:23Z  INFO    OGG-08100  Replicat added.

OGG (http://localhost:7802 dep1) 7> edit params REP2_CDC
2025-08-13T18:04:31Z  INFO    OGG-10183  Parameter file REP2_CDC.prm passed validity check.

OGG (http://localhost:7802 dep1) 8> view params REP2_CDC
REPLICAT REP2_CDC
USERIDALIAS RPDB1 DOMAIN OracleGoldenGate
REPERROR default, discard
MAP ORCLPDB1.SCOTT.EMP_E4, TARGET RPDB1.SCOTT.EMP_E4;
```

Check Status of Distribution Path on Source

```
OGG (http://localhost:7802 dep1) 9> info all
Program     Status      Group       Type              Lag at Chkpt       Time Since Chkpt

REPLICAT    STOPPED     REP2_CDC    NONINTEGRATED         00:00:00           00:01:42
REPLICAT    STOPPED     REP2_IL     NONINTEGRATED         00:00:00           00:02:56


OGG (http://localhost:7802 dep1) 10> Info replicat REP2_IL

Replicat   REP2_IL   Initialized  2025-08-14 04:02   Status STOPPED
Checkpoint Lag       00:00:00 (updated 00:03:53 ago)
Log Read Checkpoint  File ma2ma/aug14/initial_load/il000000000
                     First Record  RBA 0
Encryption Profile   LocalWallet

OGG (http://localhost:7802 dep1) 11> info replicat REP2_CDC

Replicat   REP2_CDC  Initialized  2025-08-14 04:03   Status STOPPED
Checkpoint Lag       00:00:00 (updated 00:02:49 ago)
Log Read Checkpoint  File ma2ma/aug14/cdc/ex000000000
                     First Record  RBA 0
Encryption Profile   LocalWallet
```

Start Initial Load Extract (EXT2_IL)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 32> start extract EXT2_IL
2025-08-13T18:07:00Z  INFO    OGG-00975  Extract group EXT2_IL starting.
2025-08-13T18:07:00Z  INFO    OGG-15426  Extract group EXT2_IL started.
```

Start Distribution Path for Initial Load (DSP2_IL)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 36> start DISTPATH DSP2_IL
2025-08-13T18:08:48Z  INFO    OGG-08513  The path 'DSP2_IL' has been started.
```

Start Replicat for Initial Load (REP2_IL)

```
OGG (http://localhost:7802 dep1) 12> start REPLICAT REP2_IL
2025-08-13T18:12:43Z  INFO    OGG-00975  Replicat group REP2_IL starting.
2025-08-13T18:12:43Z  INFO    OGG-15445  Replicat group REP2_IL started.
```

Check table records counts are matching both Source and Target

---

```
SCOTT@ORCLPDB1 > select count(*) from scott.emp_e4;    → Source

  COUNT(*)
----------
        15

SCOTT@RPDB1 > select count(*) from scott.emp_e4; → Target

  COUNT(*)
----------
        15
```

Start CDC Extract (EXT2_CDC)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 55> start EXT2_CDC
2025-08-13T18:30:04Z  INFO    OGG-00975  Extract group EXT2_CDC starting.
2025-08-13T18:30:04Z  INFO    OGG-15426  Extract group EXT2_CDC started.
```

Start Distribution Path for CDC (DSP2_CDC)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 58> start DISTPATH DSP2_CDC
2025-08-13T18:32:00Z  INFO    OGG-08513  The path 'DSP2_CDC' has been started.
```

Stop Distribution Path for Initial Load on Source (DSP2_IL)

```
OGG (http://localhost:7802 dep1 as ORCL_SOURCE@ORCL/CDB$ROOT) 60> stop DISTPATH DSP2_IL
2025-08-13T18:40:40Z  INFO    OGG-08514  The path 'DSP2_IL' has been stopped.
```

Stop Replicat REP2_IL on Target (Initial Load)

```
OGG (http://localhost:7802 dep1) 18> stop replicat REP2_IL
2025-08-13T18:21:51Z  INFO    OGG-08100  Sending STOP request to Replicat group REP2_IL.
2025-08-13T18:21:51Z  INFO    OGG-02965  Replicat group REP2_IL is down (gracefully).
```

Start CDC Replicat (REP2_CDC) on Target using "AFTERCSN"

```
OGG (http://localhost:7802 dep1) 20> start replicat REP2_CDC aftercsn 5945163
2025-08-13T18:33:16Z  INFO    OGG-00975  Replicat group REP2_CDC starting.
2025-08-13T18:33:16Z  INFO    OGG-15445  Replicat group REP2_CDC started.
```

Do some DML Operations on Source

```
SCOTT@ORCLPDB1 > INSERT INTO emp_e4 (EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO) VALUES
(9999, 'SHAM', 'CLERK', 7902, TO_DATE('17-DEC-1980', 'DD-MON-YYYY'), 800, NULL, 20);
1 row created.

SCOTT@ORCLPDB1 > UPDATE emp_e4  SET SAL = 900  WHERE EMPNO = 9999;
 1 row updated.

SCOTT@ORCLPDB1 > DELETE FROM emp_e4  WHERE EMPNO = 9999;
1 row deleted.

SCOTT@ORCLPDB1 > commit;
Commit complete.

OGG (http://localhost:7802 dep1) 21> Info replicat REP2_CDC, detail

Replicat   REP2_CDC  Last Started 2025-08-14 04:33   Status RUNNING
Checkpoint Lag       00:00:00 (updated 00:00:00 ago)
Process ID           77706
Log Read Checkpoint  File ma2ma/aug14/cdc/ex000000000
                     2025-08-14 04:32:41.019731  RBA 94841
Encryption Profile   LocalWallet

Current Log BSN value: (requires database login)

Last Committed Transaction CSN value: (requires database login)
```

Prepared and explored by Thiyagu Gunasekaran

```
  Extract Source                          Begin             End
  ma2ma/aug14/cdc/ex000000000             * Initialized *   2025-08-14 04:32:41.01
  ma2ma/aug14/cdc/ex000000000             * Initialized *   First Record

Current directory    /

Report file          /u01/app/oracle/product/23ai/ogg23ai_deploy/var/lib/report/REP2_CDC.rpt
Parameter file       /u01/app/oracle/product/23ai/ogg23ai_deploy/etc/conf/ogg/REP2_CDC.prm
Checkpoint file      /u01/app/oracle/product/23ai/ogg23ai_deploy/var/lib/checkpt/REP2_CDC.cpr
Checkpoint table     c##ggadmin.ckpt
Process file         /u01/app/oracle/product/23ai/ogg23ai_deploy/var/run/REP2_CDC.pcr
Error log            /u01/app/oracle/product/23ai/ogg23ai_deploy/var/log/ggserr.log
```

Check DML Operations Stats on the Target DB Server

```
OGG (http://localhost:7802 dep1) 23> stats replicat REP2_CDC

Sending STATS request to Replicat group REP2_CDC ...

Start of statistics at 2025-08-14 04:38:03.

Replicating from ORCLPDB1.SCOTT.EMP_E4 to RPDB1.SCOTT.EMP_E4:

*** Total statistics since 2025-08-14 04:35:38 ***
    Total inserts                                1.00
    Total updates                                1.00
    Total deletes                                1.00
    Total upserts                                0.00
    Total discards                               0.00
    Total operations                             3.00

*** Daily statistics since 2025-08-14 04:35:38 ***
    Total inserts                                1.00
    Total updates                                1.00
    Total deletes                                1.00
    Total upserts                                0.00
    Total discards                               0.00
    Total operations                             3.00

*** Hourly statistics since 2025-08-14 04:35:38 ***
    Total inserts                                1.00
    Total updates                                1.00
    Total deletes                                1.00
    Total upserts                                0.00
    Total discards                               0.00
    Total operations                             3.00
```

```
*** Latest statistics since 2025-08-14 04:35:38 ***
    Total inserts                                1.00
    Total updates                                1.00
    Total deletes                                1.00
    Total upserts                                0.00
    Total discards                               0.00
    Total operations                             3.00

End of statistics.

OGG (http://localhost:7802 dep1) 24>
```